

Pro A vyvůdíme singulární rozklad ze cvičení 11.7, tedy:

$$A = V \cdot D^T \cdot U^T = \begin{pmatrix} -\frac{1}{2} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & -\frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{2} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{pmatrix} \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & \sqrt{3} \end{pmatrix} \begin{pmatrix} 0 & -\frac{1}{\sqrt{3}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & -\frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & -\frac{1}{\sqrt{6}} \\ 1 & 0 & 0 & 0 \\ 0 & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{pmatrix}$$

Povůdíme nápoředu a nalezneme E takovouto, řě:

$$E \cdot D^T \cdot E = I; D^T E D^T = D^T \text{ a } E D^T = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} D^T \cdot E = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \text{ tedy:}$$

$$E = \begin{pmatrix} \frac{1}{3} & 0 & 0 \\ 0 & \frac{1}{3} & 0 \\ 0 & 0 & \frac{1}{\sqrt{3}} \\ 0 & 0 & 0 \end{pmatrix}$$

polořim $B = U E V^T$, polořim: $ABA = V D^T U^T U E V^T V D^T U^T = V D^T E D^T U^T = V D^T U^T = A$
 $BAB = U E V^T V D^T U^T U E V^T = U E D^T E V^T = U E V^T = B$

a ověřim, řě $A \cdot B$ a $B \cdot A$ jsou symetrické

$$B = \begin{pmatrix} 0 & 0 & 1 & 0 \\ -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & 0 & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & 0 & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & 0 & \frac{1}{\sqrt{3}} \end{pmatrix} \begin{pmatrix} \frac{1}{3} & 0 & 0 \\ 0 & \frac{1}{3} & 0 \\ 0 & 0 & \frac{1}{\sqrt{3}} \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{6}} & -\frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{pmatrix} = \begin{pmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{9} & \frac{1}{9} & -\frac{2}{9} \\ \frac{1}{9} & -\frac{2}{9} & \frac{1}{9} \\ -\frac{2}{9} & \frac{1}{9} & \frac{1}{9} \end{pmatrix}$$

$$A \cdot B = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{2}{3} & -\frac{1}{3} & -\frac{1}{3} \\ 0 & -\frac{1}{3} & \frac{2}{3} & -\frac{1}{3} \\ 0 & -\frac{1}{3} & -\frac{1}{3} & \frac{2}{3} \end{pmatrix}$$

$$B \cdot A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \text{ tedy } B \text{ plněně rořědky rořědky}$$